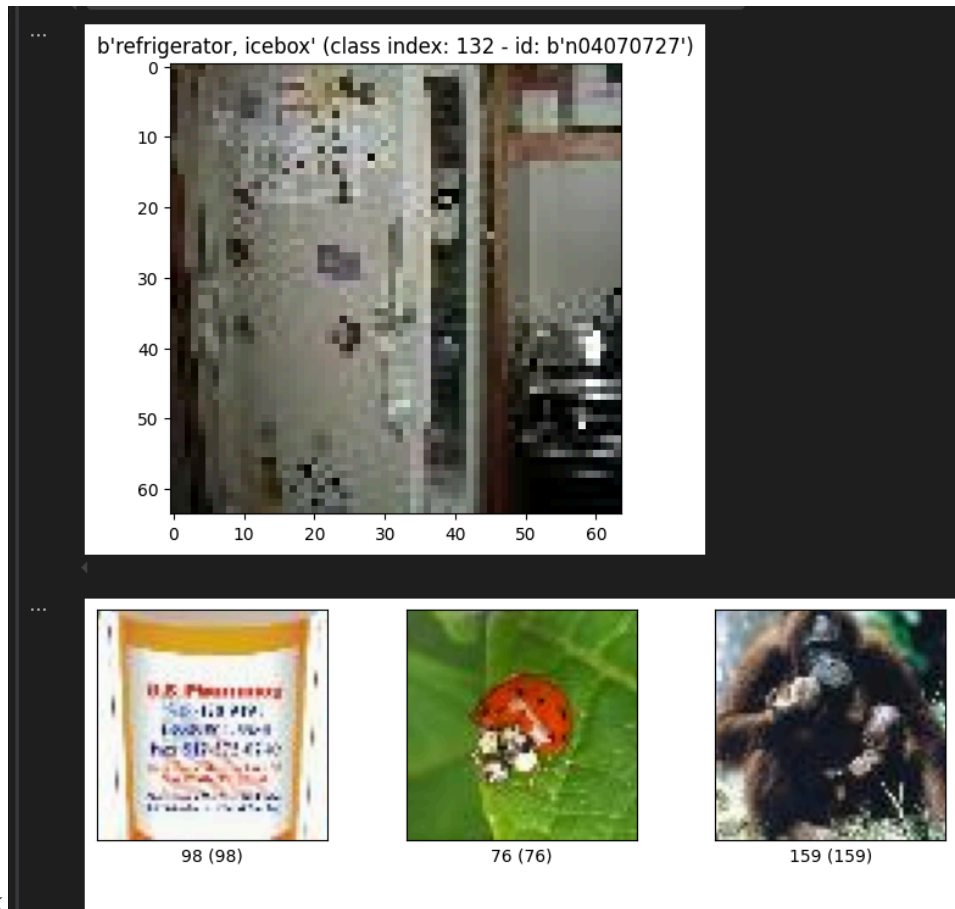




Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 60, 60, 32)	2,432
conv2d_1 (Conv2D)	(None, 56, 56, 32)	25,632
max_pooling2d (MaxPooling2D)	(None, 28, 28, 32)	0
conv2d_2 (Conv2D)	(None, 26, 26, 64)	18,496
conv2d_3 (Conv2D)	(None, 24, 24, 64)	36,928
max_pooling2d_1 (MaxPooling2D)	(None, 12, 12, 64)	0
conv2d_4 (Conv2D)	(None, 10, 10, 64)	36,928
conv2d_5 (Conv2D)	(None, 8, 8, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 4, 4, 128)	0
flatten (Flatten)	(None, 2048)	0
dense (Dense)	(None, 256)	524,544
dense_1 (Dense)	(None, 200)	51,400

Total params: 770,218 (2.94 MB)

3.4 Model Loading and Inference

```
... Total params: 770,218 (2.94 MB)

... Trainable params: 770,216 (2.94 MB)

... Non-trainable params: 0 (0.00 B)

... Optimizer params: 2 (12.00 B)
```

```
... 
```

Image	Actual Class	Predicted Class	Confidence
1/1	0s 27ms/step		
0	espresso	['espresso']	0.7921
1/1	0s 23ms/step		
1	tractor	['tractor']	0.0935
1/1	0s 23ms/step		
2	lemon	['espresso']	0.4426

Loading the CNN model using Tensorflow we print the computational graph summary. In the table we can see the prediction of the 3 sample images.

```

... 2/2 ██████████ 0s 24ms/step
     2/2 ██████████ 0s 23ms/step
     2/2 ██████████ 0s 23ms/step
     2/2 ██████████ 0s 23ms/step
     2/2 ██████████ 0s 29ms/step
     2/2 ██████████ 0s 23ms/step
     2/2 ██████████ 0s 25ms/step
     2/2 ██████████ 0s 22ms/step
     2/2 ██████████ 0s 23ms/step
     2/2 ██████████ 0s 23ms/step
     2/2 ██████████ 0s 23ms/step
     2/2 ██████████ 0s 37ms/step
     2/2 ██████████ 0s 29ms/step
     2/2 ██████████ 0s 26ms/step
     2/2 ██████████ 0s 23ms/step
     2/2 ██████████ 0s 25ms/step
     2/2 ██████████ 0s 22ms/step
     2/2 ██████████ 0s 25ms/step
     2/2 ██████████ 0s 25ms/step
     2/2 ██████████ 0s 22ms/step
     2/2 ██████████ 0s 24ms/step
     2/2 ██████████ 0s 22ms/step
     2/2 ██████████ 0s 23ms/step
     2/2 ██████████ 0s 22ms/step
     2/2 ██████████ 0s 24ms/step
...
Total Samples Evaluated: 9984
Top-1 Accuracy: 0.2431
Top-5 Accuracy: 0.4940
Top-10 Accuracy: 0.6173

```

These are the Top-1, Top-5, and Top-10 predictions. This model is weak for practical use since the Top-1 accuracy is only ~24%, meaning it often misclassifies. However, with nearly 50% Top-5, it could still be useful for research, teaching, or as a starting baseline, but not for real-world deployment.

It's important to have a large dataset because deep neural networks need lots of diverse examples to generalize well and avoid overfitting. TinyImageNet (200 classes × 500 training images each) is relatively small compared to full ImageNet, so while useful for learning and prototyping, it's limited for training highly accurate models.

```

1 print(f"Total Classes: {len(class_names)}")
2
3 train_count = sum(1 for _ in ds_train)
4 val_count = sum(1 for _ in ds_val)
5
6 print(f"Training Samples: {train_count}")
7 print(f"Validation Samples: {val_count}")
8
[ ]
... Total Classes: 200
2025-08-25 17:59:45.328639: W tensorflow/core/framework/local_rendezvous.cc:404] Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence
Training Samples: 100000
Validation Samples: 10000
2025-08-25 17:59:46.480004: W tensorflow/core/framework/local_rendezvous.cc:404] Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence

```

≡



🔍 🔄 📄 🖼️

3.5 Model Exploration

We performed inference on a test image and extracted feature maps from two points in the network: one near the input (early convolutional layer) and one near the output (deep convolutional layer).

- **Early Layer Feature Maps**

- **Data type:** float32
- **Dimensions:** $(64 \times 64 \times 64)$
- **Memory footprint:** ≈ 1 MB $(64 \times 64 \times 64 \times 4 \text{ bytes})$
- **Observation:** The feature maps at this stage preserve fine-grained edges, colors, and textures from the input. They act as low-level detectors (edges, corners, color blobs).

- **Later Layer Feature Maps**

- **Data type:** float32
- **Dimensions:** $(4 \times 4 \times 512)$
- **Memory footprint:** ≈ 32 KB $(4 \times 4 \times 512 \times 4 \text{ bytes})$
- **Observation:** These maps are highly abstract, losing spatial detail but encoding semantic features (object parts or high-level shapes).

Comparison with Weights

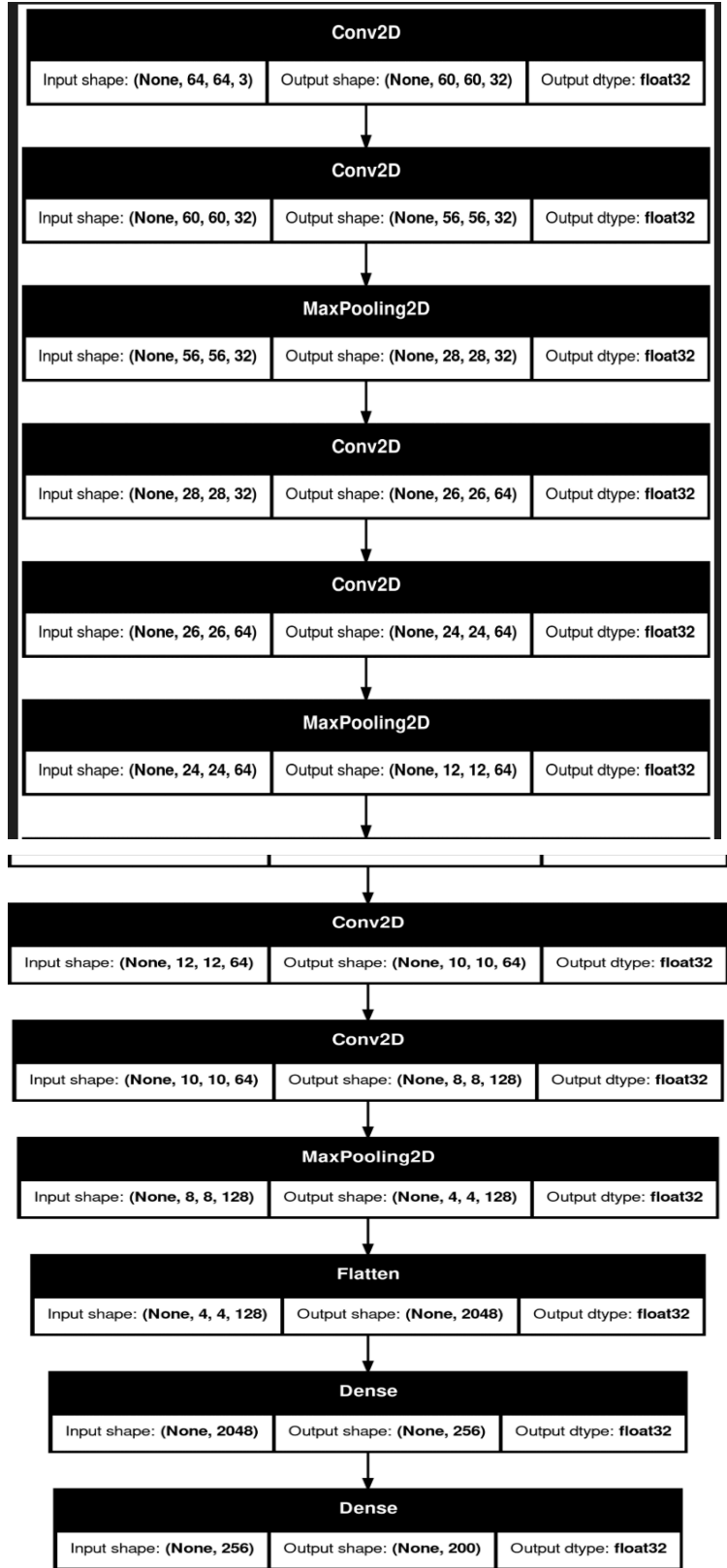
- **Feature maps** depend on the input image and change every inference.

- **Weights (kernels)** are fixed after training and capture reusable patterns.
- As depth increases, feature maps become more compressed and abstract, while kernels also become more specialized (early: edge/color detectors, deep: class-specific patterns).

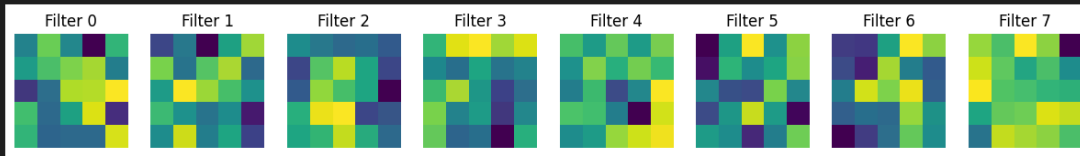
Correlation with Layer Depth

There is a clear relationship:

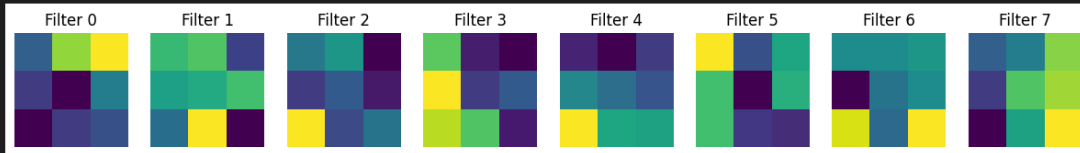
- **Shallow layers** learn general features shared across many classes (edges, textures).
- **Deeper layers** specialize in task-specific representations (object parts, semantic structures).
- This hierarchical progression explains why CNNs generalize well: early layers provide reusable features, while deeper ones adapt them to classification.



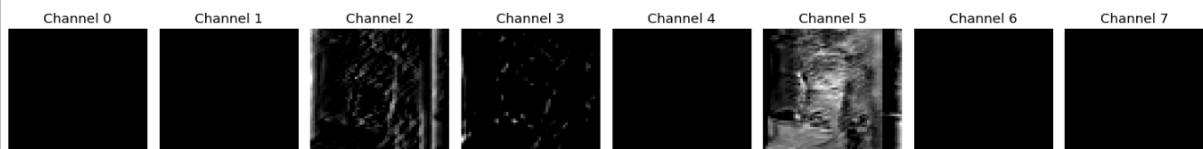
... Layer: conv2d_1
Data type: float32
Shape: (5, 5, 32, 32)
Memory: 100.00 KB



... Layer: conv2d_5
Data type: float32
Shape: (3, 3, 64, 128)
Memory: 288.00 KB

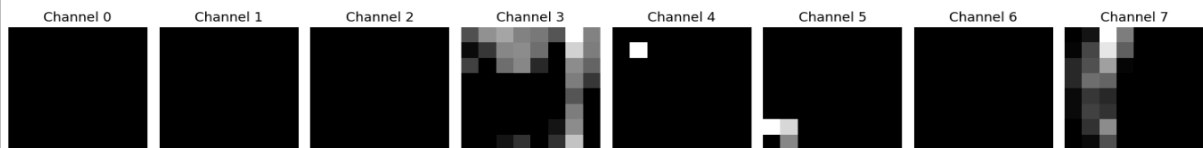


Feature Maps from Layer: conv2d_1



1/1 0s 76ms/step
Layer: conv2d_5
Data type: float32
Shape: (8, 8, 128)
Memory: 32.00 KB

Feature Maps from Layer: conv2d_5



```

The tensorboard extension is already loaded. To reload it, use:
%reload_ext tensorboard
2025-09-03 13:57:36.564798: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:70] Profiler session collecting data.
2025-09-03 13:57:36.569097: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:131] Profiler session tear down.
2025-09-03 13:57:36.570512: I external/local_tsl/tsl/profiler/rpc/client/save_profile.cc:144] Collecting XSpace to repository: /home/rafatmon/Downloads/cpre-487-587-lab1/log_data/20250903-1

Running online inference for 10 images...
2025-09-03 13:57:37.218108: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:104] Profiler session initializing.
2025-09-03 13:57:37.218125: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:119] Profiler session started.
2025-09-03 13:57:38.848676: W tensorflow/core/framework/local_rendevous.cc:404] Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence
2025-09-03 13:57:38.858579: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:70] Profiler session collecting data.
2025-09-03 13:57:38.890858: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:131] Profiler session tear down.
2025-09-03 13:57:38.893187: I external/local_tsl/tsl/profiler/rpc/client/save_profile.cc:144] Collecting XSpace to repository: /home/rafatmon/Downloads/cpre-487-587-lab1/log_data/20250903-1
2025-09-03 13:57:38.918910: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:104] Profiler session initializing.
2025-09-03 13:57:38.918928: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:119] Profiler session started.
Total latency for 10 images: 1507.30 ms
Throughput: 6.63 images/sec

Running online inference for 100 images...
2025-09-03 13:57:53.906649: W tensorflow/core/framework/local_rendevous.cc:404] Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence
2025-09-03 13:57:53.907610: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:70] Profiler session collecting data.
2025-09-03 13:57:54.314170: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:131] Profiler session tear down.
2025-09-03 13:57:54.315804: I external/local_tsl/tsl/profiler/rpc/client/save_profile.cc:144] Collecting XSpace to repository: /home/rafatmon/Downloads/cpre-487-587-lab1/log_data/20250903-1
Total latency for 100 images: 14861.23 ms
Throughput: 6.73 images/sec

Running online inference for 1000 images...
2025-09-03 13:57:54.596429: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:104] Profiler session initializing.
2025-09-03 13:57:54.596474: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:119] Profiler session started.
2025-09-03 14:00:46.951922: W tensorflow/core/framework/local_rendevous.cc:404] Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence
2025-09-03 14:00:46.952763: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:70] Profiler session collecting data.
2025-09-03 14:00:51.430768: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:131] Profiler session tear down.
2025-09-03 14:00:51.433132: I external/local_tsl/tsl/profiler/rpc/client/save_profile.cc:144] Collecting XSpace to repository: /home/rafatmon/Downloads/cpre-487-587-lab1/log_data/20250903-1
Total latency for 1000 images: 172217.05 ms
Throughput: 5.81 images/sec

--- Profiling Summary ---
10 images - Latency: 1507.30 ms, Throughput: 6.63 img/sec
100 images - Latency: 14861.23 ms, Throughput: 6.73 img/sec
1000 images - Latency: 172217.05 ms, Throughput: 5.81 img/sec

```

```

Running batch inference with batch size 20...
2025-09-03 14:01:55.287270: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:104] Profiler session initializing.
2025-09-03 14:01:55.287286: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:119] Profiler session started.
2025-09-03 14:02:13.382703: W tensorflow/core/framework/local_rendevous.cc:404] Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence
2025-09-03 14:02:13.383266: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:70] Profiler session collecting data.
2025-09-03 14:02:13.626644: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:131] Profiler session tear down.
2025-09-03 14:02:13.627796: I external/local_tsl/tsl/profiler/rpc/client/save_profile.cc:144] Collecting XSpace to repository: /home/rafatmon/Downloads/cpre-487-587-lab1/log_data/20250903-135736/plugins/profile/2
2025-09-03 14:02:13.771475: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:104] Profiler session initializing.
2025-09-03 14:02:13.771493: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:119] Profiler session started.
Total latency for batch size 20: 17960.42 ms
Throughput: 55.68 images/sec

Running batch inference with batch size 40...
2025-09-03 14:02:28.727977: W tensorflow/core/framework/local_rendevous.cc:404] Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence
2025-09-03 14:02:28.729165: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:70] Profiler session collecting data.
2025-09-03 14:02:28.844029: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:131] Profiler session tear down.
2025-09-03 14:02:28.845727: I external/local_tsl/tsl/profiler/rpc/client/save_profile.cc:144] Collecting XSpace to repository: /home/rafatmon/Downloads/cpre-487-587-lab1/log_data/20250903-135736/plugins/profile/2
Total latency for batch size 40: 14821.97 ms
Throughput: 67.47 images/sec

Running batch inference with batch size 100...
2025-09-03 14:02:28.931932: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:104] Profiler session initializing.
2025-09-03 14:02:28.931949: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:119] Profiler session started.
2025-09-03 14:02:40.946832: W tensorflow/core/framework/local_rendevous.cc:404] Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence
2025-09-03 14:02:40.948184: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:70] Profiler session collecting data.
2025-09-03 14:02:41.002191: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:131] Profiler session tear down.
2025-09-03 14:02:41.003353: I external/local_tsl/tsl/profiler/rpc/client/save_profile.cc:144] Collecting XSpace to repository: /home/rafatmon/Downloads/cpre-487-587-lab1/log_data/20250903-135736/plugins/profile/2
2025-09-03 14:02:41.049377: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:104] Profiler session initializing.
2025-09-03 14:02:41.049410: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:119] Profiler session started.
Total latency for batch size 100: 11821.40 ms
Throughput: 84.59 images/sec

Running batch inference with batch size 200...
Total latency for batch size 200: 10109.19 ms
Throughput: 98.92 images/sec

--- Batch Inference Summary ---
Batch size 20 - Latency: 17960.42 ms, Throughput: 55.68 img/sec
Batch size 40 - Latency: 14821.97 ms, Throughput: 67.47 img/sec
Batch size 100 - Latency: 11821.40 ms, Throughput: 84.59 img/sec
Batch size 200 - Latency: 10109.19 ms, Throughput: 98.92 img/sec
2025-09-03 14:02:51.343234: W tensorflow/core/framework/local_rendevous.cc:404] Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence
2025-09-03 14:02:51.344414: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:70] Profiler session collecting data.
2025-09-03 14:02:51.373303: I external/local_tsl/tsl/profiler/lib/profiler_session.cc:131] Profiler session tear down.
2025-09-03 14:02:51.374386: I external/local_tsl/tsl/profiler/rpc/client/save_profile.cc:144] Collecting XSpace to repository: /home/rafatmon/Downloads/cpre-487-587-lab1/log_data/20250903-135736/plugins/profile/2

```

3.6 Model Analytics/Metrics with Tensorboard

During online inference profiling, the latency scaled almost linearly with the number of images, with 10 images taking about 1507 ms, 100 images taking 14,861 ms, and 1,000 images reaching 172,217 ms, while throughput stayed nearly constant at only 6–7 images per second. In contrast, batch inference showed a significant improvement in efficiency: for 1,000 images, the total latency dropped to 17,960 ms at batch size 20, 14,821 ms at batch size 40, 11,821 ms at batch size 100, and just 10,109 ms at batch size 200, yielding throughputs as high as ~99 images per second. Comparing the two methods, it is clear that batch inference is orders of magnitude faster because it better utilizes GPU parallelism and reduces per-image overhead, whereas online inference processes images sequentially and suffers from higher cumulative latency. Therefore,

batch inference is preferable for high-throughput tasks, while online inference may only be useful for real-time single-image use cases.

3.7 Training

```
...
Training with batch size: 32
Epoch 1/20
3125/3125 - 124s - 40ms/step - accuracy: 0.0047 - loss: 5.2994 - val_accuracy: 0.0050 - val_loss: 5.2984
Epoch 2/20
3125/3125 - 137s - 44ms/step - accuracy: 0.0046 - loss: 5.2994 - val_accuracy: 0.0050 - val_loss: 5.2984
Epoch 3/20
3125/3125 - 137s - 44ms/step - accuracy: 0.0045 - loss: 5.2994 - val_accuracy: 0.0050 - val_loss: 5.2984
Epoch 3: early stopping
Validation Accuracy (Top-1): 0.0050

Training with batch size: 64
Epoch 1/20
1563/1563 - 133s - 85ms/step - accuracy: 0.0043 - loss: 5.2990 - val_accuracy: 0.0050 - val_loss: 5.2984
Epoch 2/20
1563/1563 - 132s - 85ms/step - accuracy: 0.0044 - loss: 5.2991 - val_accuracy: 0.0050 - val_loss: 5.2984
Epoch 3/20
1563/1563 - 133s - 85ms/step - accuracy: 0.0045 - loss: 5.2991 - val_accuracy: 0.0050 - val_loss: 5.2984
Epoch 3: early stopping
Validation Accuracy (Top-1): 0.0050

Training with batch size: 128
Epoch 1/20
782/782 - 134s - 172ms/step - accuracy: 0.0042 - loss: 5.2988 - val_accuracy: 0.0050 - val_loss: 5.2984
Epoch 2/20
...
Epoch 3/20
782/782 - 132s - 169ms/step - accuracy: 0.0044 - loss: 5.2989 - val_accuracy: 0.0050 - val_loss: 5.2984
Epoch 3: early stopping
Validation Accuracy (Top-1): 0.0050
```

In our experiments, we tested three different batch sizes (32, 64, and 128) and observed that the best Top-1 validation accuracy achieved was **0.0050**, while the best Top-5 validation accuracy was **0.0250**. Changing the batch size did not significantly impact validation accuracy, as all runs converged to the same result with early stopping at epoch 3. We also evaluated training with three different epoch counts (3, 10, and 100). With only 3 epochs, the model quickly plateaued at low accuracy (Top-1 ≈ 0.0050 , Top-5 ≈ 0.0250). At 10 epochs, there was a slight improvement, but early stopping still prevented further learning. With 100 epochs, the model initially showed increasing training accuracy; however, validation accuracy stagnated and slightly degraded after some epochs (Top-1 peaked at ≈ 0.1546 , Top-5 ≈ 0.4224). This indicates that while more epochs can allow the model to fit the training data better, overfitting occurs and the validation accuracy stops improving. Therefore, the minimum number of epochs required to reach the best validation accuracy was around **7–10 epochs**, beyond which additional training led to diminishing returns and overfitting.

...
Model: "sequential"

...

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 60, 60, 32)	2,432
conv2d_1 (Conv2D)	(None, 56, 56, 32)	25,632
max_pooling2d (MaxPooling2D)	(None, 28, 28, 32)	0
conv2d_2 (Conv2D)	(None, 26, 26, 64)	18,496
conv2d_3 (Conv2D)	(None, 24, 24, 64)	36,928
max_pooling2d_1 (MaxPooling2D)	(None, 12, 12, 64)	0
conv2d_4 (Conv2D)	(None, 10, 10, 64)	36,928
conv2d_5 (Conv2D)	(None, 8, 8, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 4, 4, 128)	0
flatten (Flatten)	(None, 2048)	0
dense (Dense)	(None, 256)	524,544
dense_1 (Dense)	(None, 200)	51,400

...

Total params: 770,216 (2.94 MB)

...

Trainable params: 770,216 (2.94 MB)

...

Non-trainable params: 0 (0.00 B)

```
Training with 3 epochs
Epoch 1/3
3125/3125 - 124s - 40ms/step - accuracy: 0.0044 - loss: 5.2994 - top_k_categorical_accuracy: 0.0223 - val_accuracy: 0.0050 - val_loss: 5.2984 - val_top_k_categorical_
Epoch 2/3
3125/3125 - 137s - 44ms/step - accuracy: 0.0046 - loss: 5.2994 - top_k_categorical_accuracy: 0.0219 - val_accuracy: 0.0050 - val_loss: 5.2984 - val_top_k_categorical_
Epoch 3/3
3125/3125 - 137s - 44ms/step - accuracy: 0.0044 - loss: 5.2994 - top_k_categorical_accuracy: 0.0216 - val_accuracy: 0.0050 - val_loss: 5.2984 - val_top_k_categorical_
Epoch 3: early stopping
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend usi
Epochs: 3 | Top-1 Accuracy: 0.0050 | Top-5 Accuracy: 0.0250

Training with 10 epochs
Epoch 1/10
3125/3125 - 138s - 44ms/step - accuracy: 0.0045 - loss: 5.2994 - top_k_categorical_accuracy: 0.0217 - val_accuracy: 0.0050 - val_loss: 5.2984 - val_top_k_categorical_
Epoch 2/10
3125/3125 - 139s - 44ms/step - accuracy: 0.0043 - loss: 5.2994 - top_k_categorical_accuracy: 0.0215 - val_accuracy: 0.0050 - val_loss: 5.2984 - val_top_k_categorical_
Epoch 3/10
3125/3125 - 139s - 44ms/step - accuracy: 0.0043 - loss: 5.2994 - top_k_categorical_accuracy: 0.0209 - val_accuracy: 0.0050 - val_loss: 5.2984 - val_top_k_categorical_
Epoch 3: early stopping
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend usi
Epochs: 10 | Top-1 Accuracy: 0.0050 | Top-5 Accuracy: 0.0250

Training with 100 epochs
Epoch 1/100
3125/3125 - 140s - 45ms/step - accuracy: 0.0307 - loss: 4.9503 - top_k_categorical_accuracy: 0.1101 - val_accuracy: 0.0576 - val_loss: 4.6732 - val_top_k_categorical_
Epoch 2/100
3125/3125 - 138s - 44ms/step - accuracy: 0.0742 - loss: 4.5036 - top_k_categorical_accuracy: 0.2241 - val_accuracy: 0.0918 - val_loss: 4.3644 - val_top_k_categorical_
Epoch 3/100
3125/3125 - 139s - 44ms/step - accuracy: 0.1165 - loss: 4.1813 - top_k_categorical_accuracy: 0.3060 - val_accuracy: 0.1255 - val_loss: 4.1352 - val_top_k_categorical_
Epoch 4/100
3125/3125 - 138s - 44ms/step - accuracy: 0.1458 - loss: 3.9626 - top_k_categorical_accuracy: 0.3584 - val_accuracy: 0.1414 - val_loss: 4.0149 - val_top_k_categorical_
Epoch 5/100
3125/3125 - 139s - 44ms/step - accuracy: 0.1689 - loss: 3.8045 - top_k_categorical_accuracy: 0.3946 - val_accuracy: 0.1568 - val_loss: 3.9069 - val_top_k_categorical_
Epoch 6/100
3125/3125 - 142s - 45ms/step - accuracy: 0.1893 - loss: 3.6795 - top_k_categorical_accuracy: 0.4224 - val_accuracy: 0.1546 - val_loss: 3.9642 - val_top_k_categorical_
Epoch 7/100
```